Journal First Novelty Statement

Adaptive HACT: Information Limits and Safe Adaptation of Verification Evidence

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The manuscript presents new research on reorganizing verification evidence without changing the acceptance contract. It proves that completion distributions with identical singleton and pairwise probabilities can have disjoint globally optimal certificate trees, and establishes sharp minimax regret across admissible affine packet prices and fanouts. A second construction separates two layouts under fixed batches of eight. The paper gives a conditional publication invariant for migration guarded by generation numbers and an exact global reference for exchangeable complete batches, specializing established tree optimization for batch updates.

These results address the first Journal First criterion through representations of joint completions under a protected registry. They address the second criterion because the central information limit and safety composition are substantive results, not merely additional proofs, routine algorithm variants, or extra experiments appended to a prior publication. A conditional adoption rule incorporates calibration, distribution shift, and installation cost. The evaluation includes controlled workload changes and comparisons against ledgers with maintained counters, retaining unfavorable cases rather than claiming a general service or agent speedup.

Earlier drafts, code, and evidence are publicly archived in the project repository. The paper cites established work on tree optimization, hypergraphs, and verification frameworks and does not claim those ingredients as new.

Statement body: 200 words.